

Y12 Mathematics Extension 1 | Term 1 Assessment 2012

Question 1 (9 Marks)

Marks

(a) Find:

(i) $\int e^{2x-1} dx$ 1

(ii) $\int_0^1 \sqrt{1-x^2} dx$, where $\theta = \sin^{-1} x$. 3

(b) Find the exact value of the gradient to the curve $y = \operatorname{cosec} x$ at $x = \frac{5\pi}{6}$ 2

(c) If $f'(x) = 2 \cot x - x$ and $f\left(\frac{\pi}{2}\right) = 0$, find an expression for $f(x)$ 3

Question 2 (9 Marks) – START A NEW PAGE

Marks

(a) (i) Find the sum of the series S given: 3

$$S = 2^n + 2^{n-1} + 2^{n-2} + \dots + 2^{-2n}$$

(ii) Discuss the existence of a limiting sum as $n \rightarrow \infty$ for S , giving reasons. 1

(b) Write $\sqrt{6} \sin x + \sqrt{2} \cos x$ in the form $R \cos(x + \alpha)$, 3
where $R > 0$ and $0 < \alpha < 2\pi$

(c) Find k , if $x^{k+3} = e^{7 \ln x}$, where $x > 0$ 2

Question 3 (9 Marks) – START A NEW PAGE**Marks**

- (a) A spherical bubble is expanding so that its volume increases at a constant rate of 40mm^3 per second. What is the rate of increase of the surface area when the radius is 10 mm? **3**

- (b) Prove by mathematical induction that:

$$1 + 4 + 16 + \dots + 4^n = \frac{1}{3}(4^{n+1} - 1), \text{ for } n = 0, 1, 2, \dots \quad \mathbf{3}$$

- (c) Determine all values of x for which **3**
 $\log_5(x - 2) + \log_5(x - 6) = 1$

Question 4 (9 Marks) – START A NEW PAGE**Marks**

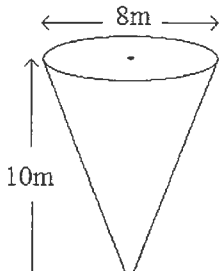
- (a) (i) Sketch the graph of $y = x \ln x$, showing all important features. **3**

- (ii) Explain the nature of the curve as $x \rightarrow 0$, giving reasons. **1**

- (b) (i) Show that $\frac{d}{dx}(\tan^3 x) = 3 \sec^4 x - 3 \sec^2 x$. **2**

- (ii) Using (i) or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \sec^4 x \, dx$. **3**

- (a) When the same constant k is added to each of the numbers 60, 100, and 150 respectively, a geometric sequence is obtained. Find the common ratio for the sequence formed. 3

- (b)  A large grain storage container is in the shape of an inverted cone. The diameter of the top is 8 metres. The vertical height is 10 metres.

- (i) If h (in metres) is the height of the grain at any given time, show that the volume V cubic metres at the time is given by $V = \frac{4}{75}\pi h^3$. 2
- (ii) Grain runs out at the bottom at the rate of 3 cubic metres per second. 4
Find the rate of change of the height of grain in the container at the instant when the height is 5 metres. Give your answer in exact form.

- (a) Use the principle of mathematical induction to prove $3^n > 2n + 4$, for $n = 2, 3, 4, \dots$. 3
- (b) Andrew takes out a loan for \$50,000 to buy a new car. The interest on the loan is 9% per annum, compounded monthly. He agrees to pay the loan by equal monthly instalments over 5 years.
- (i) Show that each monthly repayment will be approximately \$1 037.92. 3
- (ii) Assuming he repays \$1 037.92 per month, how much is still owing after making the 30th repayment? 2
- (iii) After the 30th repayment he makes a lump sum payment of \$10,000. If Andrew wishes to still make 30 more repayments, what would be the new value of each repayment? 1

- (a) A flashlight throws a cone of light with a 60° angle between the outermost rays and the axis of the cone. A man points the light straight at a wall. At what rate is the illuminated area of the wall changing at the moment when the light is 3 metres from the wall and is being brought towards the wall at a rate of 0.25 metre per second? 3
- (b) A sector of angle θ in radians is cut from a circular disc of radius 4π cm and used to make the complete curved surface of a right circular cone (with no overlap).
- (i) Prove that the volume of this cone is given by: 3
- $$V = \frac{8\pi\theta^2}{3} \sqrt{4\pi^2 - \theta^2}$$
- (ii) Find the value of θ for which the volume of this cone is a maximum. 3

END OF PAPER

1/2

MATHEMATICS Extension 1 : Question...1....		
Suggested Solutions	Marks	Marker's Comments
(a) (i) $\int e^{2x-1} dx = \frac{1}{2} e^{2x} + C$ (ii) $\int_0^{\pi/2} \sqrt{1-x^2} dx$ $= \int_0^{\pi/2} \sqrt{1-\sin^2 \theta} \cdot \cos \theta d\theta$ $= \int_0^{\pi/2} \cos^2 \theta d\theta$ $= \frac{1}{2} \int_0^{\pi/2} (\cos 2\theta + 1) d\theta$ $= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\pi/2}$ $= \frac{1}{2} [0 + \pi/2 - 0]$ $= \pi/4$	1 1/2 for limits 1/2 1/2 1/2 1/2 1/2	lost 1/2 a mark if forgot "+C"
or Area = 1/4 of unit circle $= \frac{1}{4} \pi \times 1^2$ $= \pi/4$	1 1 1	If the radius = 1 wasn't mentioned, then full marks wasn't awarded.
b) $y = \operatorname{cosec} x$ $\frac{dy}{dx} = \frac{-\cos x}{\sin^2 x}$ $= -\cot x \cdot \operatorname{cosec} x$ when $x = 5\pi/6$ $m = \frac{-\cos 5\pi/6}{\sin^2(5\pi/6)}$ $= \frac{\sqrt{3}/2}{1/4}$ $= 2\sqrt{3}$ $\therefore$ gradient is $2\sqrt{3}$	(1) 1/2 1/2	

2/2

MATHEMATICS Extension 1 : Question 1

Suggested Solutions

Marks

Marker's Comments

(c) $f'(x) = 2\cot x - x$

$f(x) = 2\ln(\sin x) - \frac{1}{2}x^2 + C$

when $x = \frac{\pi}{2}$ $f(x) = 0$

$0 = 2\ln(\sin \frac{\pi}{2}) - \frac{1}{2}(\frac{\pi}{2})^2 + C$

$0 = 0 - \frac{\pi^2}{8} + C$

$C = \frac{\pi^2}{8}$

$\therefore f(x) = 2\ln(\sin x) - \frac{x^2}{2} + \frac{\pi^2}{8}$

3

1/2 of each error!

Suggested Solutions	Marks	Marker's Comments
a2) $S = 2^n + 2^{n-1} + 2^{n-2} + \dots + 2^{-2n}$ ① $\frac{1}{2}S = 2^{n-1} + 2^{n-2} + \dots + 2^{-2n-1}$ ② $S - \frac{1}{2}S = 2^n - 2^{-2n-1}$ $\frac{1}{2}S = 2^n - 2^{-2n-1}$ $S = 2^{n+1} - 2^{-2n}$ # or $a = 2^n$ $r = \frac{1}{2}$ # of terms = $3n+1$ $S = \frac{2^n [1 - (\frac{1}{2})^{3n+1}]}{1 - \frac{1}{2}}$ $S = 2^{n+1} (1 - 2^{-3n-1})$ $S = 2^{n+1} - 2^{-2n}$ # aii) as $n \rightarrow \infty$ $2^{n+1} \rightarrow \infty$, $2^{-2n} \rightarrow 0$ $S \rightarrow \infty$ $\therefore$ no limiting sum		1m for $3n+1$ many students made mistake in # of terms or mixed up the 'n' in formula $S_n = \frac{a(r^n - 1)}{r - 1}$ Calculation will be made easier next, 1 1/2 m
b) $\sqrt{6} \sin x + \sqrt{2} \cos x = R(\cos x \cos d - \sin x \sin d)$ $R \sin d = -\sqrt{6}$ $R \cos d = \sqrt{2}$ ($R > 0$) $R = \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} = 2\sqrt{2}$ $\tan d = -\sqrt{3}$ in 4th Q as $\sin d < 0$, $\cos d > 0$ $d = 5\frac{\pi}{3}$ $2\sqrt{2} \cos(x + 5\frac{\pi}{3})$ #	1 m 1 m 1 m	aii) must give correct reason for no limiting sum. $r < 1$ or $r > 1$ $\therefore$ irrelevant here forgot $2\sqrt{2} \cos(x + 5\frac{\pi}{3})$ - 1/2 m
c) $x^{k+3} = e^{(k+3)\ln x} = x^7$ $k = 4$	1 m 1 m	

MATHEMATICS Extension 1 : Question 3

Suggested Solutions	Marks	Marker's Comments
a) Given $\frac{dv}{dt} = 40$ $V = \frac{4}{3} \pi r^3 \quad \therefore \frac{dv}{dr} = 4\pi r^2$ $\therefore \frac{dv}{dr} = \frac{dv}{dt} \times \frac{dr}{dt}$ $4\pi r^2 = 40 \times \frac{dr}{dt}$ $\therefore \frac{dr}{dt} = \frac{10}{\pi r^2}$ $A = 4\pi r^2 \quad \therefore \frac{dA}{dr} = 8\pi r$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ $= 8\pi r \times \frac{10}{\pi r^2}$ $= \frac{80}{r}$ When $r = 10$ $\frac{dA}{dr} = 8$ Surface area is increasing at $8 \text{ mm}^2/\text{sec}$	(3)	(1) $\frac{dr}{dt}$ (1) $\frac{dA}{dt}$ ($\frac{1}{2}$) correct answer with units ($\frac{1}{2}$) answering the question

MATHEMATICS Extension 1 : Question 3

Suggested Solutions

Marks

Marker's Comments

b) Let $P(n)$ be the proposition that:
 $1 + 4 + 16 + \dots + 4^n = \frac{1}{3}(4^{n+1} - 1)$ for
 $n = 0, 1, 2, \dots$

Test $P(0)$: LHS = 1

$$RHS = \frac{1}{3}(4^1 - 1) = \frac{3}{3} = 1$$

$\therefore$ LHS = RHS $\therefore P(0)$ is true

Assume $P(k)$ is true for some $k = 0, 1, 2, \dots$

ie $1 + 4 + 16 + \dots + 4^k = \frac{1}{3}(4^{k+1} - 1)$

Required to Prove $P(k+1)$ is true

ie $1 + 4 + 16 + \dots + 4^{k+1} = \frac{1}{3}(4^{k+2} - 1)$

$$\begin{aligned} \text{LHS} &= 1 + 4 + 16 + \dots + 4^k + 4^{k+1} \\ &= \frac{1}{3}(4^{k+1} - 1) + 4^{k+1} \text{ (by assumption)} \\ &= \frac{1}{3}(4^{k+1}) - \frac{1}{3} \\ &= \frac{1}{3}[4 \times 4^{k+1} - 1] \\ &= \frac{1}{3}[4^{k+2} - 1] \\ &= \text{RHS.} \end{aligned}$$

$\therefore P(n)$ is true by principle of mathematical induction

(3)

① Must test $n=0$.
 must show substitution for RHS.

} ②

"by assumption"
 ② substitution

} ① algebra

no marks for conclusion if incorrect working

MATHEMATICS Extension 1 : Question.....3

Suggested Solutions	Marks	Marker's Comments
c) $\log_5(x-2) + \log_5(x-6) = 1$ $\log_5[(x-2)(x-6)] = 1$ $\therefore x^2 - 8x - 12 = 5$ $\therefore x^2 - 8x - 7 = 0$ $(x-7)(x+1) = 0$ $\therefore x = 7 \text{ or } x = -1$ But $x > 6 \quad \therefore x \neq -1$ $\therefore x = 7 \text{ (only)}$	(3)	. (1/2) combine logs (1/2) multiply out ① Solve quadratic for 2 solutions ① reject $x = -1$.

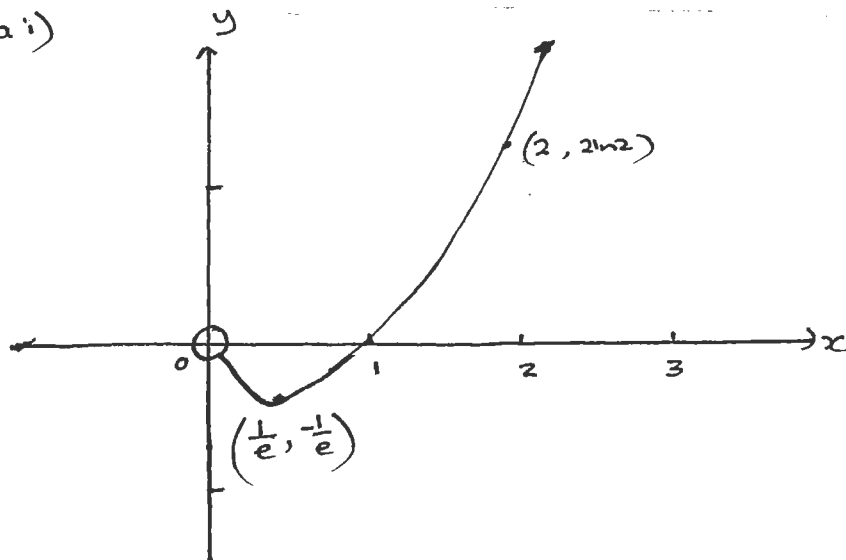
MATHEMATICS Extension 1 : Question 4

Suggested Solutions

Marks

Marker's Comments

4a i)



Calculus $y = x \ln x$

x int put $y = 0$

$$x \ln x = 0$$

$$x = 1$$

stationary points

$$y' = \ln x + 1$$

$$\ln x + 1 = 0$$

$$\ln x = -1$$

$$x = e^{-1}$$

$$\therefore y = \frac{1}{e} \ln e = -\frac{1}{e}$$

$$\therefore \text{SP is } \left(\frac{1}{e}, -\frac{1}{e}\right)$$

Test nature $y'' = \frac{1}{x}$

$$\text{at } x = e^{-1} \quad y'' = \frac{1}{e^{-1}}$$

$$= e$$

> 0 concave up

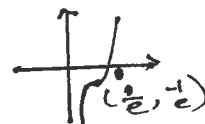
absolute minimum point at $\left(\frac{1}{e}, -\frac{1}{e}\right)$

3 marks

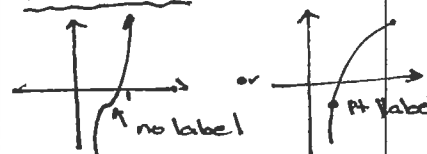
Marks were awarded for :

- x-int
- shape
- open circle at zero
- TP showing $\left(\frac{1}{e}, -\frac{1}{e}\right)$
- Scale on y-axis.

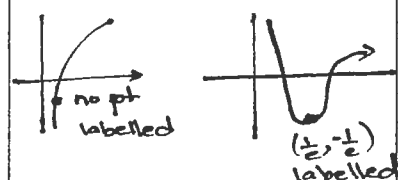
2 marks



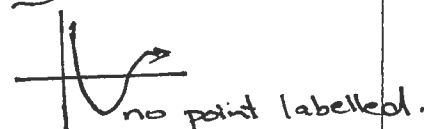
1 1/2 marks



1 mark



1/2 mark



ii) Question says explain

As $x \rightarrow 0$, the gradient becomes negatively very steep (the curve approaches (0,0) with a negative gradient. It approaches a vertical tangent.

1 mark

No marks for statement formulae only

- you had to mention gradient

- No CFE because one had to go back to the original question to look at gradients.

4

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
4b(i) $\frac{d}{dx} \tan^3 x = 3 \tan^2 x \cdot \sec^2 x \checkmark$ $= 3(\sec^2 x - 1) \sec^2 x \checkmark$ $= 3 \sec^4 x - 3 \sec^2 x.$	2 marks	Some used Quotient rule $\tan^3 x = \frac{\sin^3 x}{\cos^3 x}.$ * Be careful with writing superscripts eg $\tan^3 x$ looked like $\tan^3 x$
From (i) $\int_0^{\pi/4} \frac{d}{dx} [\tan^3 x] = 3 \int_0^{\pi/4} \sec^4 x \, dx - 3 \int_0^{\pi/4} \sec^2 x \, dx \checkmark$ $\int_0^{\pi/4} \sec^4 x \, dx = \frac{1}{3} \int_0^{\pi/4} \frac{d}{dx} (\tan^3 x) + \int_0^{\pi/4} \sec^2 x \, dx$ $= \frac{1}{3} [\tan^3 x]_0^{\pi/4} + [\tan x]_0^{\pi/4} \checkmark$ $= \frac{1}{3} (1 - 0) + (1 - 0)$ $= \frac{4}{3} \checkmark$ $\therefore \int_0^{\pi/4} \sec^4 x \, dx = \frac{4}{3}$	3 marks	Some students got $\frac{2}{3}$ because they divided both $\tan^3 x$ and $\tan x$ by 3 $\frac{1}{3} (\tan^3 x + \tan x)$ 0.

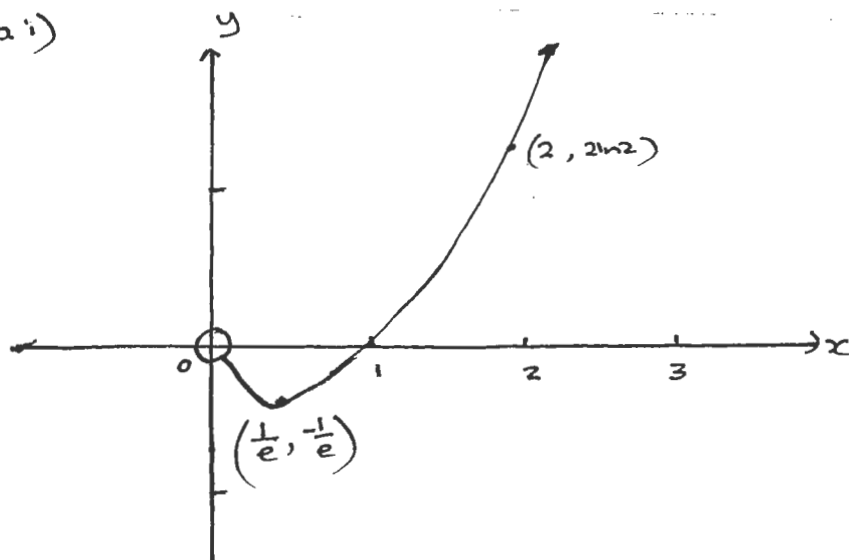
MATHEMATICS Extension 1 : Question 4

Suggested Solutions

Marks

Marker's Comments

4a i)



Calculus $y = x \ln x$

x int put $y = 0$

$x \ln x = 0$

$x = 1$

stationary points

$y' = \ln x + 1$

$\ln x + 1 = 0$

$\ln x = -1$

$x = e^{-1}$

$\therefore y = \frac{1}{e} \ln e = -\frac{1}{e}$

$\therefore$ SP is $(\frac{1}{e}, -\frac{1}{e})$

Test nature $y'' = \frac{1}{x}$

at $x = e^{-1}$ $y'' = \frac{1}{e^{-1}}$

$= e > 0$ concave up

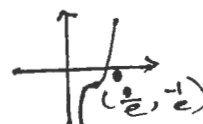
absolute minimum point at $(\frac{1}{e}, -\frac{1}{e})$

3 marks

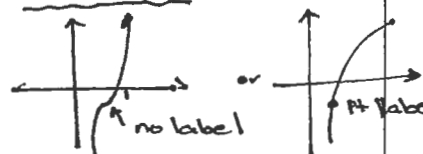
Marks were awarded for:

- x-int
- shape
- open circle at zero
- TP showing $(\frac{1}{e}, -\frac{1}{e})$
- Scale on y-axis.

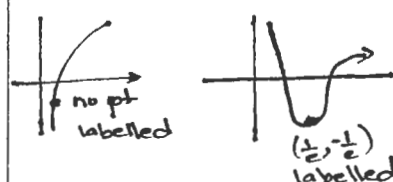
2 marks



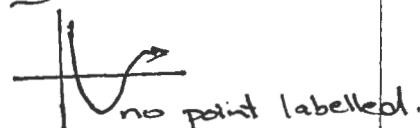
1/2 marks



1 mark



1/2 mark



ii) Question says explain

As $x \rightarrow 0$, the gradient becomes negatively very steep (the curve approaches (0,0) with a negative gradient. It approaches a vertical tangent.

1 mark

No marks for statement formulae only

- you had to mention gradient

• No CFE because one had to go back to the original question to look at gradients.

4
MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
4b(i) $\frac{d}{dx} \tan^3 x = 3 \tan^2 x \cdot \sec^2 x \checkmark$ $= 3(\sec^2 x - 1) \sec^2 x \checkmark$ $= 3 \sec^4 x - 3 \sec^2 x.$	2 marks	Some used Quotient rule $\tan^3 x = \frac{\sin^3 x}{\cos^2 x}.$ * Be careful with writing superscripts eg $\tan^3 x$ looked like $\tan^3 x$
From (i) $\int_0^{\pi/4} \frac{d}{dx} [\tan^3 x] = 3 \int_0^{\pi/4} \sec^4 x \, dx - 3 \int_0^{\pi/4} \sec^2 x \, dx \checkmark$ $\int_0^{\pi/4} \sec^4 x \, dx = \frac{1}{3} \int_0^{\pi/4} \frac{d}{dx} (\tan^3 x) + \int_0^{\pi/4} \sec^2 x \, dx$ $= \frac{1}{3} [\tan^3 x]_0^{\pi/4} + [\tan x]_0^{\pi/4} \checkmark$ $= \frac{1}{3} (1-0) + (1-0)$ $= \frac{4}{3}$ $\therefore \int_0^{\pi/4} \sec^4 x \, dx = \frac{4}{3} \checkmark$	3 marks	Some students got $\frac{2}{3}$ because they divided both $\tan^3 x$ and $\tan x$ by 3 $\frac{1}{3} (\tan^3 x + \tan x)$ 0.

MATHEMATICS Extension 1 : Question ... 5

Suggested Solutions

Marks

Marker's Comments

a) The amended numbers are in geometric sequence

$$\frac{100+k}{60+k} = \frac{150+k}{100+k} = r$$

$$(100+k)^2 = (150+k)(60+k)$$

$$10000 + 200k + k^2 = 9000 + 210k + k^2$$

$$10k = 1000$$

$$k = 100$$

$$r = \frac{100+100}{60+100} = \frac{5}{4}$$

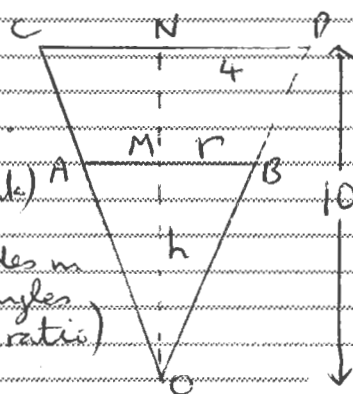
b) i) Let point O, A, B, C, D, M and N be as shown.

$\triangle OMB \parallel \triangle ONP$ (Equiangular)

$$\frac{h}{10} = \frac{r}{4} \quad (\text{Correspond sides in similar triangles are in the same ratio})$$

$$r = \frac{2h}{5}$$

$$V_{\text{old}} = V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75} h^3$$



I knocked 1/2 mark if no mention of similar triangles or equivalent (tan etc)

There was no mark for just knowing the formula for V.

c) $\frac{dV}{dt} = -3$ (decreasing)

$$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$$

$$\text{But } \frac{dh}{dV} = 1/\frac{dV}{dh} = 1/\frac{12h^2\pi}{75} = \frac{25}{4h^2\pi}$$

$$\frac{dh}{dt} = -3 \times \frac{25}{4h^2\pi} = -\frac{3 \times 25}{4\pi \times 5^2} \quad \text{when } h=5$$

$$= -\frac{3}{4\pi}$$

Grain level is decreasing (going down) at $\frac{3}{4\pi}$ m/s

Chain Rule, explicit or direct.

Calculus

Answer

Interpretation of - signs. (lots of) (1/2 docked)

Y12 T1

MATHEMATICS Extension 1 : Question 6

Suggested Solutions

Marks

Marker's Comments

a) step 1 $n=2$ LHS = 3^2 RHS = $2 \times 2 + 4$
 $= 9$ $= 8$
 $\therefore$ LHS > RHS
 step 2 Assume statement true $n=k$ k is integer
 $\therefore 3^k > 2k+4$ OR $3^k - 2k - 4 > 0$
 To prove true $n=k+1$
 $\therefore 3^{k+1} > 2(k+1) + 4$
 $\therefore 3^{k+1} > 2k+6$ OR $3^{k+1} - 2k - 6 > 0$
 New $3^{k+1} - 2k - 6 = 3 \cdot 3^k - 2k - 6$
 $= 3[3^k - 2k - 4] + 6k + 12 - 2k - 6$
 $= 3[3^k - 2k - 4] + 4k + 6$
 > 0 by assumption
 $\therefore 4k + 6 > 0$ for $k > 2$

1

 $\frac{1}{2}$ $\frac{1}{2}$

1

$\therefore$ true $n=k$ it is also true $n=k+1$
 since true $n=2$ then it is true for $n=4, 2=3$
 $n=3, 1=4$ check 10 on for the integers 7, 2
 w) (i) A_n = amount owing, R = Repayment $r = 0.0075$
 Amount owing end of month = $50000 \times 1.0075 - A$ $A = 30$
 and month = $(50000 \times 1.0075 - R) \times 1.0075 - R$

To be done
3 times.

$A_1 = 50000 \times 1.0075 - R[1 + 1.0075]$
 End 3 months = $[50000 \times 1.0075^3 - R[1 + 1.0075 + 1.0075^2] - R]$
 $A_3 = 50000 \times 1.0075^3 - R[1 + 1.0075 + 1.0075^2]$
 $A_{60} = 0$
 $\therefore 50000 \times 1.0075^{60} - R[1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{59}]$
 $50000 \times 1.0075^{60} = R \left[\frac{1.0075^{60} - 1}{1.0075 - 1} \right]$

1

$R = \frac{50000 \times 1.0075^{60} \times 0.0075}{1.0075^{60} - 1}$
 $= 1037.9177$

1

Formula for R do
 be stated as
 multi-decimal value

(ii) $A_{20} = 50000 \times 1.0075^{20} - \left[\frac{1.0075^{30} - 1}{0.0075} \right]$

2

End 30 months \$27790.26

(iii) owing \$17790.26

$A = 17790.26 \times 1.0075^{30} \times 0.0075$
 $= \$664.43$

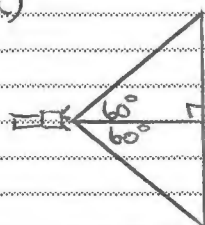
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Suggested Solutions

Marks

Marker's Comments

(a)



Let the distance from the flashlight to the wall at any instant be x m.
 $\therefore$ radius of circle of light $= x \tan 60^\circ = \sqrt{3}x$ m

The illuminated area (A) is

$$A = \pi (\sqrt{3}x)^2$$

$$A = 3\pi x^2$$

Now

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$= 6\pi x \times -0.25 \quad \text{given } \frac{dx}{dt} = -0.25$$

$$= 18\pi \times -0.25 \quad \text{when } x = 3$$

$$= -9\pi \text{ m}^2/\text{s}$$

$\therefore$ The illuminated area is decreasing at the rate of $9\pi \text{ m}^2/\text{s}$ when the light is 3m from the wall.

①

An answer of $\frac{\pi}{2} \text{ m}^2/\text{s}$ received max of 2 marks only

Needed $\frac{dx}{dt} = -0.25$ and the final sentence area is decreasing to gain a full mark

①

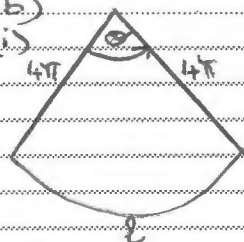
Wrong units - $\frac{1}{2}$ mark

①

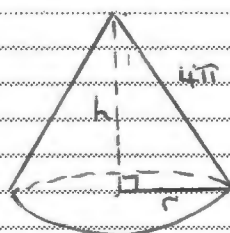
Having written decreasing at the rate of $-9\pi \text{ m}^2/\text{s}$ lost $\frac{1}{2}$ mark

(b)

(i)



$$l = 4\pi\theta$$



$$l = 4\pi\theta \quad \text{and} \quad 2\pi r = 4\pi\theta$$

$$r = 2\theta$$

Now

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi \cdot 4\theta^2 \cdot \sqrt{16\pi^2 - R^2}$$

$$V = \frac{4}{3}\pi\theta^2 \sqrt{16\pi^2 - 4\theta^2}$$

$$V = \frac{4}{3}\pi\theta^2 \times 2\sqrt{4\pi^2 - \theta^2}$$

$$\therefore V = \frac{8\pi\theta^2}{3} \sqrt{4\pi^2 - \theta^2}$$

①

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Generally well done

MATHEMATICS Extension 1 : Question 7...

Suggested Solutions

Marks

Marker's Comments

(b)
(ii)

$$V = \frac{8\pi\theta^2}{3} \sqrt{4\pi^2 - \theta^2}$$

$$\frac{dV}{d\theta} = \frac{8\pi\theta^2}{3} \times \frac{1}{2\sqrt{4\pi^2 - \theta^2}} \times (-2\theta) + \sqrt{4\pi^2 - \theta^2} \times \frac{16\pi\theta}{3}$$

$$\frac{dV}{d\theta} = \frac{8\pi\theta}{3} \left[\frac{8\pi^2 - 3\theta^2}{\sqrt{4\pi^2 - \theta^2}} \right]$$

For stationary pts $\frac{dV}{d\theta} = 0$

$$\frac{dV}{d\theta} = 0 \text{ when } 3\theta^2 = 8\pi^2 \text{ or } \theta = 0$$

$\theta = 0$ is rejected since $\theta > 0$

$$\therefore \theta = \pm \frac{2\sqrt{6}\pi}{3} = \pm \frac{2\sqrt{6}\pi}{3}$$

but $0 < \theta < 2\pi$

$$\therefore \theta = \frac{2\sqrt{6}\pi}{3} \quad (\approx 5.13^\circ = 294^\circ)$$

Test nature

θ	$\frac{\sqrt{6}\pi}{3}$	$\frac{2\sqrt{6}\pi}{3}$	$\sqrt{6}\pi$
$\frac{dV}{d\theta}$	6π	0	-10π
	> 0		< 0

Since $V = f(\theta)$ is continuous for all θ in $0 < \theta < 2\pi$ and $\frac{dV}{d\theta} = 0$ has one value in

the domain, then there is an absolute maximum value at $\theta = \frac{2\sqrt{6}\pi}{3}$

①

1 mark for correct derivative

①

$\frac{1}{2}$ mark deducted if $\theta = 0$ was factored out and not given as a soln

①

$\frac{1}{2}$ mark deduction for not considering $\theta = -\frac{2\sqrt{6}\pi}{3}$ and

discounting it as $0 < \theta < 2\pi$

If students used the second derivative test they needed to show the second derivative and the value when $\theta = \frac{2\sqrt{6}\pi}{3}$ is substituted into $\frac{d^2V}{d\theta^2}$